

## Multiple Choice (10 marks)

1. Which of the following is the source of oxygen produced during photosynthesis?
  - (a) O<sub>2</sub>
  - (b) CH<sub>4</sub> (Methane)
  - (c) CO<sub>2</sub>
  - (d) H<sub>2</sub>O
  - (e) N<sub>2</sub> (Nitrogen)
2. Why is genetic drift less important in human evolution at present than it was twenty thousand or more years ago?
  - (a) The human genome has fully adapted to current environmental pressures
  - (b) Modern medicine has eliminated the effects of genetic drift
  - (c) The rate of genetic drift is directly proportional to technological advancements
  - (d) Humans have reached a genetic equilibrium where drift is no longer possible
  - (e) Humans have developed immunity to genetic drift
3. Briefly discuss some abnormalities of the respiratory tract.
  - (a) Influenza, chickenpox, and measles
  - (b) Multiple sclerosis, Parkinson's disease, and Alzheimer's disease
  - (c) Arthritis, osteoporosis, and fibromyalgia
  - (d) Hypertension, heart disease, and diabetes
  - (e) Chronic obstructive pulmonary disease (COPD), cystic fibrosis, and pleural effusion
4. Crossing-over during meiosis permits scientists to determine
  - (a) the rate of mutations
  - (b) the chance for variation in zygotes
  - (c) the age of an organism
  - (d) the potential for genetic disorders
  - (e) the sequence of DNA bases
5. The parathyroid glands in man are small, pealike organs. They are usually four in number and are located on the surface of the thyroid. What is the function of these glands?

- (a) The parathyroid glands produce insulin to regulate blood sugar levels
- (b) The parathyroid glands regulate blood pressure
- (c) The parathyroid glands control breathing rate
- (d) The parathyroid glands secrete enzymes that break down proteins
- (e) The parathyroid glands control the balance of electrolytes in the body

## **True / False (10 marks)**

*Answer True or False*

- 6. True or False: An HIV 1/2 point of care test on sputum for screening TB/HIV co-infection in Central India - Will it work?
- 7. True or False: Does the type of tibial component affect mechanical alignment in uni-compartmental knee replacement?
- 8. True or False: Is Acupuncture Efficacious for Treating Phonotraumatic Vocal Pathologies?
- 9. True or False: Can elevated troponin I levels predict complicated clinical course and inhospital mortality in patients with acute pulmonary embolism?
- 10. True or False: Does a febrile reaction to platelets predispose recipients to red blood cell alloimmunization?

## **Long Form Response (20 marks)**

*Answer each question with a clear and complete explanation.*

- 11. Discuss the concept of multiple alleles in genetics, their origin, and their role in controlling various traits in a population.
- 12. Discuss why conjugation is unlikely to occur among paramecia derived from a single individual through repeated fission, focusing on genetic identity and its implications for genetic exchange.

*End of Paper*